Achieving Covert Communications in Ultra-Dense LEO Satellite Systems by Exploiting Interference and Directional Uncertainty

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***Abstract*—This paper investigates a multi-satellite cooperative covert satellite communication (SatCom) scheme, where a positive covert rate is achieved in ultra-dense low Earth orbit (LEO) satellite constellations by exploiting both interference and directional uncertainty. Specifically, interference arises from aggregate sidelobe leakage from other satellite transmissions, while directional uncertainty stems from the random selection of the transmitting satellite among multiple accessible ones. To this end, we first propose a LEO satellite network model and formulate the corresponding hypothesis testing problem for the cooperative covert SatCom scheme. The power distribution of the aggregate interference is quantified and approximated using stochastic geometry. Next, by analyzing the detection error probability and outage probability under the impact of aggregate interference, we derive an approximate covert capacity expression for the case of a single accessible satellite, which maintains a positive covert rate even as the slot length approaches infinity. Furthermore, by leveraging directional uncertainty through hiding the signal's angle of arrival, we analyze the multi-satellite cooperative covert SatCom scheme, leading to a concise approximate expression that reveals significant covert capacity improvement. Numerical simulations are performed to verify the superiority of the proposed scheme, suggesting that a positive covert capacity can be achieved with interference uncertainty and significantly enhanced by directional uncertainty as the number of satellites increases.**

***Index Terms*—Covert communication, satellite communications, ultra-dense LEO satellite system, interference uncertainty, directional uncertainty.**

I. Introduction

A. Motivations

SINCE the inception of satellite communication (SatCom), it has played a pivotal role in global telecommunications, supporting a range of applications such as radio broadcasting, weather forecasting, and navigation [1]. Recently, standardization efforts have underscored SatCom's critical role in enabling the upcoming sixth-generation (6G) networks [2]. As internet-based applications continue to expand, the focus of SatCom system design has increasingly shifted toward data services, particularly broadband SatCom. Reflecting this trend, several commercial entities, including SpaceX and OneWeb, have launched large low Earth orbit (LEO) satellite constellations aimed at delivering satellite-based cellular services [3].

However, due to the broadcast nature of wireless channels, SatCom signals are susceptible to interception by any device within the coverage area, posing significant risks to personal privacy and information security [4]. Traditional security measures typically rely on message encryption [5], which involves creating complex passwords and encryption schemes to protect confidentiality, or on physical layer security [6], [7], which exploits wireless channels to safeguard signal integrity. However, these methods do not prevent the detection of the transmission itself, and encrypted messages may even invite further probing by eavesdroppers. Additionally, the positions of transmitting nodes are often exposed, heightening the risk of jamming attacks [8]. In such cases, if the communication signal remains undetected by eavesdroppers, the satellite system can then transition to a higher level of wireless security.

In the context of SatCom security, covert communication, also known as low probability of detection (LPD) communication, has emerged as a promising solution to render wireless transmissions undetectable [9], [10], [11]. Significant efforts have been made to integrate covert communication into SatCom systems. For example, transmission strategies in space-air-ground integrated vehicular networks have been explored [9], along with beamformer and trajectory design for unmanned aerial vehicles (UAVs) in uplink SatCom [10]. Additionally, the covert outage probability of a dual-hop space-air-ground integrated network has been considered for finite block-length communications [11]. However, as demonstrated in Bash's seminal work [12], up to $O(\sqrt{n})$ bits can be transmitted over an additive white Gaussian noise (AWGN) channel in n channel uses, under arbitrarily low detection probabilities. This result suggests a pessimistic outlook for covert SatCom, as the covert capacity approaches zero asymptotically.

B. Related Works

The pessimistic zero covert capacity implied by the above square-root law has spurred research into the possibility of

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achieving a positive covert rate. By exploiting the signal-to-noise ratio (SNR) wall caused by uncertain noise power [13], it is possible to achieve a positive asymptotic covert capacity [14]. Subsequent studies have shown that uncertainty from extrinsic interference can also induce the SNR wall phenomenon, enabling a positive asymptotic covert capacity [15], [16], [17], [18]. For instance, Hu proposed a covert communication scheme with a positive effective covert capacity, leveraging the uncertainty of artificial noise generated by a full-duplex receiver [15]. In [16], a UAV was employed as a jammer to enhance covert communication by using artificial interference, with the covert rate maximization problem formulated as a Nash bargaining game. Liu [17] investigated the impact of aggregate interference from other transmitters, demonstrating the potential of exploiting uncertain extrinsic interference. In [18], covert throughput in an Internet-of-Things (IoT) network with dense deployment was derived, considering both background noise and aggregate interference from other devices.

Inspired by covert schemes based on interference uncertainty, covert SatCom has also attracted significant research attention [19], [20], [21]. A stochastic optimization problem has been formulated in covert SatCom systems, to maximize the minimum covert rate under communication constraints [19]. By randomizing the variance of transmitted Gaussian signals, a covert SatCom scheme with a positive rate was designed [20]. Feng [21] proposed a two-stage Stackelberg game to model the conflict dynamics between terrestrial base stations and large-scale multi-tier LEO satellite networks, optimizing covert SatCom performance. In summary, existing works demonstrate that interference, whether artificially generated or ambient, can increase the uncertainty of the eavesdropper's received signal, impairing its ability to detect the transmission, and thereby enhancing the capacity of covert SatCom systems.

C. Contributions

Although covert SatCom has been extensively studied, the inherent characteristics of ultra-dense LEO satellite constellations present new opportunities for designing advanced covert SatCom schemes. On one hand, sidelobe leakages from other satellites' data transmissions can generate aggregate interference without requiring additional resources for artificial interference [22]. On the other hand, due to the directional nature of satellite communication, downlink signals from different transmitting satellites arrive at varying angles of arrival.

Inspired by these characteristics, we propose a novel covert SatCom scheme that leverages both the uncertainty in aggregate interference power and directional uncertainty to conceal communication signals. Specifically, by exploiting the beam-hopping technique [23] in large-scale LEO constellations, the proposed scheme enables ground users to be dynamically served by satellites from different directions across time slots in an aggregate interference environment, thereby jointly exploiting interference power and directional uncertainty. We then derive approximate expressions for the covert capacity of SatCom schemes with both single and multiple accessible satellites, exploring how covert capacity varies with different system parameters.

The main contributions of this paper are as follows:

- A covert SatCom scheme is proposed for ultra-dense LEO satellite constellations, exploiting both aggregate interference and directional uncertainty. Multiple satellites collaboratively transmit covert signals to a legitimate receiver, where the eavesdropper faces the dual challenge of identifying the transmitting satellite from spatially dispersed candidates and discerning the signals embedded within the dynamically aggregate interference.
- A multi-tier satellite distribution model is developed using stochastic geometry and then the power distribution of aggregate interference is investigated by considering sidelobe leakage from ultra-dense LEO satellite constellations. By modeling the spatial distribution of satellites as a stochastic point process across multiple spherical shells, an expression for the aggregate interference power is derived, and its distribution is approximated.
- Approximate expressions for the covert capacity of both single-satellite and multi-satellite covert SatCom schemes are derived, and the impact of key system parameters on covert capacity is investigated. Theoretical and numerical analyses show that covert capacity of the proposed schemes remains positive as the slot length approaches infinity, and the capacity of multi-satellite schemes increases monotonically with the number of satellites.

The rest of this paper is organized as follows. Section II presents the system model and hypothesis testing. Section III introduces the satellite model and analyzes the aggregate interference. The covert capacity for the single-satellite scheme is derived in Section IV. Section V extends the analysis to the multi-satellite cooperative scheme. Section VI provides numerical results and investigates the impact of various parameters on covert capacity. Finally, Section VII concludes the paper.

II. System Model

A. Network Model

As shown in Fig. 1, a covert SatCom system is considered, which consists of a set of ultra-dense LEO satellites $\mathcal{S} = \{1, 2, \cdots, N\}$ and some active ground users including a legitimate receiver named Bob. Downlink covert transmission links are desired to be established from one accessible satellite named Alice to Bob. Meanwhile, there exists a ground warden Willie located near Bob, aiming to monitor any possible transmission to Bob. Owing to the dense deployment of LEO satellites, there is a group of satellites accessible to Bob. Such a group is denoted by the set $\mathcal{S}_A \subseteq \mathcal{S}$, whose size $N_A = \rho_A N$ and $\rho_A \in (0, 1)$ is the proportion of accessible satellites in $\mathcal{S}$.

In particular, we assume that there will be M slots of consecutive downlink covert transmissions, where for each slot $m \in \mathcal{M} = \{1, 2, \ldots, M\}$, one target accessible satellite s_m will be randomly selected from $\mathcal{S}_A$ as Alice, responsible for transmitting covert signals to Bob in the presence of Willie.

Meanwhile, other transmitting satellites in $\mathcal{S}$ will potentially cause co-frequency interference on the covert transmissions due to sidelobe leakages of their downlink signals. All these satellites are denoted by the set of interfering satellites $\mathcal{S}_I \subseteq \mathcal{S}$, whose cumulative sidelobe leakages are defined as the aggregate interference. Note that such kind of co-frequency interference will impact Bob's signal reception and Willie's

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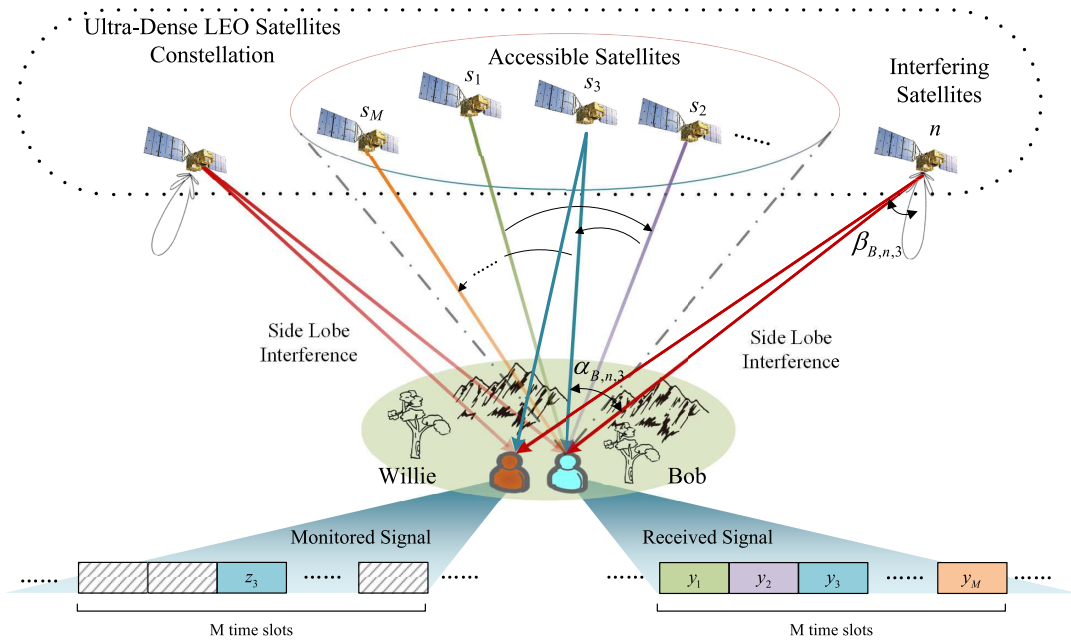

Fig. 1. The multi-satellite cooperative covert SatCom system in the ultra-dense LEO satellite scenario.

monitoring simultaneously. As for the mainlobe of satellite signals, typically their mainlobe directions should be properly coordinated by the satellite system, and thus interference from them is not considered here without loss of generality.

1) Communication Link From Alice to Bob: For each slot $m \in \mathcal{M}$, the covert signal emitted by the selected accessible satellite $s_m \in \mathcal{S}_A$ is normalized as the vector $\boldsymbol{c}_m = (c_m[1], c_m[2], \cdots, c_m[L])$, where each symbol follows a circularly symmetric complex Gaussian (CSCG) distribution, i.e., $c_m[l] \sim \mathcal{CN}(0,1)$ for $1 \le l \le L$. Thus, the receiving signal $\boldsymbol{y}_m$ of Bob can be written by,

$$\boldsymbol{y}_m = \sqrt{\frac{G_B G_A P_{s_m,m}}{r_{B,s_m,m}^{\alpha_{\mathrm{PL}}}}} h_{B,s_m,m} \boldsymbol{c}_m + \boldsymbol{v}_{B,m} + \boldsymbol{i}_{B,m}, \quad (1)$$

where G_B is the receiving antenna gain of Bob, G_A is the transmitting antenna gain of each accessible satellite in $\mathcal{S}_A$, $P_{s_m,m}$ represents the transmit power of the selected accessible satellite s_m, and $h_{B,s_m,m}$ is the small-scale fading coefficient of the link from satellite s_m to Bob in slot m. Besides, $r_{B,s_m,m}$ denotes the link distance and α_{PL} is path loss exponent. Along with the attenuated covert signal, the channel noise at Bob is written by $\boldsymbol{v}_{B,m}$, where $v_{B,m}[l] \sim \mathcal{CN}(0, \sigma_B^2)$.

More importantly, $\boldsymbol{y}_m$ also includes $\boldsymbol{i}_{B,m}$, i.e., the aggregate interference experienced by Bob[1] from $\mathcal{S}_I$, where $i_{B,m}[l] \sim \mathcal{CN}(0, I_{B,m})$. Here, the corresponding aggregate interference power $I_{B,m}$ can be calculated as follows,

$$I_{B,m} = \sum_{n \in \mathcal{S}_I} \frac{G'_B(\alpha_{B,n,m}) G'_I(\beta_{B,n,m}) P_{n,m} |h_{B,n,m}|^2}{r_{B,n,m}^{\alpha_{\mathrm{PL}}}}, \quad (2)$$

where $G'_B(\alpha_{B,n,m})$ is the sidelobe gain of Bob's receiving antenna and approaches G_B when $\alpha_{B,n,m} \to 0$, $\alpha_{B,n,m}$ denotes the off-axis angle of receiving antenna, $G'_I(\beta_{B,n,m})$ represents the transmitting antenna sidelobe gain of each interfering satellite in $\mathcal{S}_I$, and $\beta_{B,n,m}$ denotes the off-axis angle of transmitting antenna of interfering satellite n. Additionally, $P_{n,m}$ is the transmit power, $h_{B,n,m}$ is the small-scale fading coefficient, and $r_{B,n,m}$ is the link distance, for each interference link from satellite $n \in \mathcal{S}_I$ to Bob.

2) Communication Link From Alice to Willie: As for the warden Willie, he always attempts to intercept and detect any possible covert signal transmissions. Notably, due to the low receiving gain of omnidirectional antennas, it is assumed that Willie is equipped with a high-gain directional antenna, which indicates that he is capable of monitoring only one target accessible satellite, or in other words, covert signal's arrival direction, for each slot m.

In fact, since he has no knowledge of the random selection of the accessible satellite, he needs to decide which direction to monitor. In this case, a practical and reasonable monitoring strategy is to persist in a fixed target direction corresponding to an accessible satellite $s_W \in \mathcal{S}_A$ throughout all the M slots. It is worth noting that such strategy is equivalent to the random monitoring strategies with the same successful probability.

Moreover, even with adaptive monitoring, the proposed covert burst transmissions make it difficult for Willie to accumulate sufficient samples within a short time to infer the satellite selection [25]. Thus, periodic updates of the satellite selection pattern effectively counter such strategies and preserve communication covertness.

Therefore, the receiving signal $\boldsymbol{z}_m$ of Willie at slot m under the persistent monitoring strategy can be expressed as,

$$\boldsymbol{z}_m = \delta_{s_W,s_m} \sqrt{\frac{G_W G_A P_{s_m,m}}{r_{W,s_m,m}^{\alpha_{\mathrm{PL}}}}} h_{W,s_W,m} \boldsymbol{c}_m + \boldsymbol{v}_{W,m} + \boldsymbol{i}_{W,m}, \quad (3)$$

[1]The aggregate interference caused by the sidelobe leakage of multiple satellites is affected by various factors and is therefore typically assumed to follow a Gaussian distribution [24].

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where the indicator δ_{s_W,s_m} equals 1 when Willie's monitoring satellite s_W matches the selected transmitting satellite s_m in slot m and 0 otherwise. G_W is the receiving antenna gain of Willie, $h_{W,s_m,m}$ is the small-scale fading coefficient of the link from satellite s_m to Willie in each slot m, and $r_{W,s_m,m}$ is the link distance. Similar to (2), Willie also experiences the channel noise $\boldsymbol{v}_{W,m}$ and the aggregate interference $\boldsymbol{i}_{W,m}$, each symbol of which can be denoted as $v_{W,m}[i] \sim \mathcal{CN}(0,\sigma_W^2)$ and $i_{W,m}[i] \sim \mathcal{CN}(0, I_{W,m})$, respectively, for $1 \le i \le L$. Particularly, the aggregate interference power $I_{W,m}$ at Willie can be expressed as follows,

$$I_{W,m} = \sum_{n\in\mathcal{S}_I} \frac{G'_W(\alpha_{W,n,m})G'_I(\beta_{W,n,m})P_{n,m}|h_{W,n,m}|^2}{r_{W,n,m}^{\alpha_{\mathrm{PL}}}}, \tag{4}$$

where $G'_W(\alpha_{W,n,m})$ is the sidelobe gain of Willie's receiving antenna and approaches G_W when $\alpha_{W,n,m} \to 0$, $\alpha_{W,n,m}$ and $\beta_{W,n,m}$ denote the off-axis angle of receiving antenna and transmitting antenna, $h_{W,n,m}$ is the small-scale fading coefficient, and $r_{W,n,m}$ is the link distance, for each interference link from satellite $n \in \mathcal{S}_I$ to Willie.

3) Discussions: In the proposed covert SatCom system, two key sources of uncertainty can be leveraged, namely interference uncertainty and directional uncertainty. Specifically, the power of aggregate interference fluctuates across different time slots due to variations in the set of active transmitting satellites and the inherent fluctuations in the small-scale fading coefficient, which will be theoretically quantified in Section III. Additionally, the lack of knowledge about the chosen accessible satellite during each time slot presents Willie with the challenge of determining which direction to monitor. Consequently, both sources of uncertainty serve to enhance the transmission of covert satellite signals, leading to an increase in the covert transmission rate.

Moreover, in order to guarantee reliable covert transmission across all the M slots, Alice employs a channel inversion power control strategy[2] [27]. This ensures that the received signal power at Bob remains constant at $P_{B,0}$ by adjusting the satellite transmit power $P_{s_m,m}$, such that $P_{B,m} = \frac{G_B G_A P_{s_m,m}}{r_{B,s_m,m}^{\alpha_{\mathrm{PL}}}} \triangleq P_{B,0}$. Similarly, since Willie is located near Bob, the distances from each accessible satellite to both Bob and Willie are approximately equal, i.e., $r_{W,s_m,m} \approx r_{B,s_m,m} \triangleq r_{s_m,m}$ for all $s_m \in \mathcal{S}_A$. As a result, Willie's received signal power, $P_{W,m}$, also remains constant, with $P_{W,m} = \frac{G_W G_A P_{s_m,m}}{r_{s_m,m}^{\alpha_{\mathrm{PL}}}} = \frac{G_W}{G_B} P_{B,m} \triangleq P_{W,0}$. Due to the typically favorable LoS conditions of accessible satellites, their links are modeled as fading channels with constant attenuation, which will be accounted for in the above power control strategy to ensure stable received power $P_{B,0}, P_{W,0}$. Meanwhile, interference links, which are more likely to encounter multipath effects, are modeled as Rician fading channels.

B. Hypothesis Testing of Detection at Willie

To detect the existence of covert signals under interference and directional uncertainty, Willie firstly faces the challenge of identifying which satellite is transmitting throughout each of the M slots, since he is not aware of the selected satellite s_m. Beyond that, Willie also needs to separate the covert signal from the aggregate interference and noise.

In summary, during the signal transmission across all the M slots, Willie aims to distinguish between the following two hypotheses based on his received signal sequence $\{\boldsymbol{z}_m\}_{m=1}^M$,

$$\mathbf{H}_0 : \text{accessible satellites do not transmit}, \tag{5}$$

$$\mathbf{H}_1 : \text{accessible satellites transmit during the M slots}. \tag{6}$$

Note that the covert transmission is conducted over M consecutive time slots, and Willie's successful detection of the covert signal in any single slot will be considered as evidence of the accessible satellites' transmission.

Under these two hypotheses, Willie's received signal $\boldsymbol{z}_m$ during each slot m is represented as follows,

$$\boldsymbol{z}_m = \begin{cases} \boldsymbol{v}_{W,m} + \boldsymbol{i}_{W,m}, & \mathbf{H_0}, \\ \delta_{s_W,s_m}\sqrt{P_{W,0}}\boldsymbol{c}_m + \boldsymbol{v}_{W,m} + \boldsymbol{i}_{W,m}, & \mathbf{H_1}, \end{cases} \tag{7}$$

where, under hypothesis $\mathbf{H}_1$, Willie can successfully capture the attenuated covert signal if and only if the monitored satellite s_W coincides with the transmitting satellite s_m during slot m, i.e., $\delta_{s_W,s_m} = 1$. Before detection, however, Willie does not know which slots satisfy the condition $\delta_{s_W,s_m} = 1$, as the relationship between s_W and the dynamically selected s_m is unknown.

Given that a radiometer-based detection framework is generally adopted for practical detection at Willie [28], the following test statistic $T_W(M,L)$, which is the average power of the received signal $\boldsymbol{z}_m$ over M slots, is employed to perform a threshold test,

$$T_W(M,L) = \frac{1}{M}\sum_{m=1}^{M} \frac{\boldsymbol{z}_m^H \boldsymbol{z}_m}{L} \underset{D_0}{\overset{D_1}{\gtrless}} \gamma, \tag{8}$$

where γ denotes the decision threshold, and D_1 and D_0 are decisions supporting $\mathbf{H}_1$ and $\mathbf{H}_0$, respectively.

Then, we can define the probability of false alarm $P_{\mathrm{FA}} = \mathbb{P}(D_1|\mathbf{H}_0) = \mathbb{P}(T_W(M,L) > \gamma|\mathbf{H}_0)$, and the probability of misdetection $P_{\mathrm{MD}} = \mathbb{P}(D_0|\mathbf{H}_1) = \mathbb{P}(T_W(M,L) < \gamma|\mathbf{H}_1)$, respectively. Under the assumption of equal *a prior* probabilities of $\mathbf{H}_0$ and $\mathbf{H}_1$, the error detection probability of Willie's algorithm is calculated as $\eta = P_{\mathrm{FA}} + P_{\mathrm{MD}}$, where $\eta \in [0,1]$ is a function of the decision threshold γ.

Assuming that Willie is equipped with the optimal detector, he will find the appropriate detection threshold $\hat{\gamma}$ to obtain the minimum error detection probability $\eta_{\min}(M,L) = \min_\gamma(P_{\mathrm{FA}} + P_{\mathrm{MD}})$, which is shown as follows,

$$\eta_{\min}(M,L) = \min_\gamma\{\mathbb{P}(T_W(M,L) > \gamma|\mathbf{H}_0) + \mathbb{P}(T_W(M,L) < \gamma|\mathbf{H}_1)\}. \tag{9}$$

As indicated by (9), Willie's minimum detection error probability $\eta_{\min}(M,L)$ is affected by the statistical properties of the test statistic T_W, which represents the average power of the received signals. According to (7), the received sequence $\{\boldsymbol{z}_m\}_{m=1}^M$ comprises additive noise $\boldsymbol{v}_{W,m}$, aggregate interference $\boldsymbol{i}_{W,m}$, and potentially information symbols $\boldsymbol{c}_m$, with the power of both noise and information components assumed to be known. As a result, the statistical behavior of interference power $I_{W,m}$ becomes critical in the analysis of $\eta_{\min}(M,L)$.

[2]The channel inversion power control strategy adapts the transmit power of satellites across slots according to estimated channel attenuation (e.g., free-space loss, cloud attenuation, and rain attenuation), similar to the power adaptation employed in Starlink for users at different elevation angles [26]. Moreover, due to the inherently low transmit power in covert communication, the transmit power rarely approaches the satellite power constraint.

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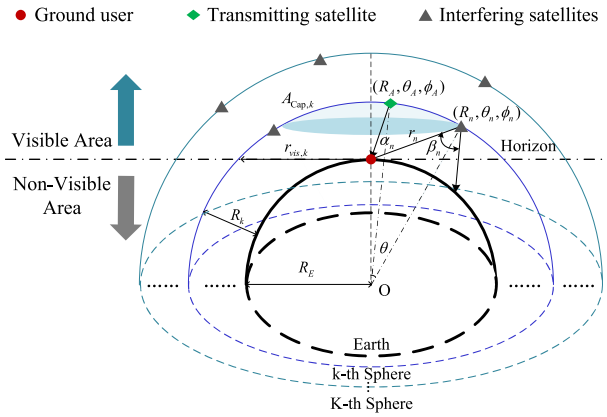

Fig. 2. A sketch of the LEO satellite system, where ground users, e.g., Bob and Willie, are located on the surface of Earth with radius R_E and satellites are distributed randomly on multiple spherical surfaces with radius $R_E + R_k$.

III. Interference Analysis in Ultra-Dense LEO Satellite Systems

Based on the system model of the proposed covert SatCom scheme, to lay the groundwork for the subsequent analysis of covert capacity subject to aggregate interference, a stochastic geometry framework is employed to model the aggregate interference and examine the statistical characteristics of its power. Specifically, by characterizing the spatial distribution of satellites as a stochastic point process on multiple spherical shells at different altitudes, we derive the expression for the aggregate interference and investigate the distributional properties of its key parameters. Furthermore, we approximate the interference power distribution using the method of moments, fitting it to a Gamma distribution.

A. Representation of Aggregate Interference

As illustrated in Fig. 2, a multi-tier satellite distribution model is developed to characterize the aggregate interference in ultra-dense LEO constellations. The constellation $\mathcal{S}$ consists of K independent orbital shells, expressed as $\mathcal{S} = \bigcup_{k=1}^{K} \mathcal{S}_k$, where $\mathcal{S}_k$ denotes the subset of satellites in the k-th shell. Satellites within each shell are uniformly distributed over a concentric spherical surface with radius $R_E + R_k$, measured from Earth's center, where R_E is the Earth's radius and R_k is the altitude of the k-th spherical surface above Earth's surface. The spatial distribution of satellites on each spherical surface follows a binomial point process (BPP) with population size $N_k = \rho_k N$, where $\rho_k \in [0, 1]$ denotes the proportion of satellites in the k-th shell and satisfies $\sum_{k=1}^{K} \rho_k = 1$.

To quantify the aggregate interference, we consider a ground user Bob located on the surface of Earth. Bob is served by and directs his antenna towards a selected accessible satellite Alice positioned at the polar coordinate (R_A, θ_A, ϕ_A). As for other satellites on the k-th spherical surface transmitting on the same frequency, each satellite $n \in \mathcal{S}_{I,k}$ positioned at (R_n, θ_n, ϕ_n) serves his respective target users. Thus, interference will be introduced by sidelobe leakage from the downlink beam, and the aggregate interference I_{ag} can be expressed as follows,

$$I_{\text{ag}} = \sum_{k=1}^{K} \sum_{n \in \mathcal{S}_{I,k}} \frac{G'_R(\alpha_n) G'_I(\beta_n) P_I |h_n|^2}{r_n^{\alpha_{\text{PL}}}}, \tag{10}$$

where $G'_R(\alpha_n)$ and $G'_I(\beta_n)$ are the sidelobe gains of the receiving and transmitting antennas, respectively. α_n and β_n denote the off-axis angles of the corresponding antenna. P_I is the transmit power of the interfering satellite, h_n is the small-scale fading coefficient, and r_n is the link distance. Note that $\mathcal{S}_{I,k}$ denotes the subset of interference satellites over the k-th spherical surface, $\mathcal{S}_I = \bigcup_{k=1}^{K} \mathcal{S}_{I,k}$.

According to ITU-R S.1528 [29], the sidelobe gain of the antenna with off-axis angle α can be calculated as

$$G'(\alpha) = \begin{cases} G_{\max}, & 0^\circ \le \alpha < \phi_b, \\ G_{\max} - 3\left(\dfrac{\alpha}{\phi_b}\right)^2, & \phi_b \le \alpha < Y, \\ G_{\max} + L_s - 25\log\left(\dfrac{\alpha}{Y}\right), & Y \le \alpha < Z, \\ L_F, & Z \le \alpha \le 180^\circ, \end{cases} \tag{11}$$

where $G_{\max}$ is the maximum gain of the antenna, $2\phi_b$ denotes the half power beamwidth of antennas, and other parameters Y, L_s, L_F, Z are empirical and are set according to values provided in ITU-R S.1528. Considering interference mitigation strategies in LEO satellite constellations, the feasible ranges of α_n, β_n are $[\alpha_{\min}, 180^\circ]$ and $[\beta_{\min}, 180^\circ]$ respectively, where $\alpha_{\min}$ and $\beta_{\min}$ denote the minimum off-axis angles of transmitting and receiving antenna.[3]

Leveraging the stationary property of BPP [31], the aggregate interference power experienced by Bob across different time slots is independent and identically distributed as the random variable I_{ag}. Additionally, since Willie is located near Bob and is equipped with a similar receiving antenna, the distances and off-axis angles from each satellite to both receivers are approximately equal. As a result, the aggregate interference experienced by Willie can be quantified in a similar manner.

B. Power Distribution Analysis of Aggregate Interference

Based on the multi-tier satellite model, the polar coordinates of any satellite on spherical shell k can be represented in the following lemma.

Lemma 1: The polar coordinates of satellites on spherical shell k can be modeled as $(R_E + R_k, \theta, \phi)$, where the elevation angle θ follows a distribution with the probability density function (PDF) $f_E(\theta) = \frac{1}{2}\sin(\theta)$ on $[0, \pi]$ and the azimuth angle ϕ follows a uniform distribution over $[-\pi, \pi]$.

Proof: Leveraging the random property of BPP, the azimuth angle ϕ follows a uniform distribution over interval $[-\pi, \pi]$. Besides, the cumulative distribution function (CDF) $F_E(\theta_0)$ of the elevation angle $\theta \le \theta_0$ from any satellite on the k-th spherical surface to ground receivers is related to the surface area of the shaded spherical cap $A_{\text{cap},k}(\theta_0)$ in Fig. 2, i.e., $F_E(\theta) = \frac{A_{\text{cap},k}(\theta)}{4\pi(R_E+R_k)^2}$. From basic geometry, $A_{\text{cap},k}(\theta) = 2\pi(R_E + R_k)^2(1 - \cos\theta)$. Then, $F_E(\theta) = \frac{1-\cos\theta}{2}, \theta \in [0, \pi]$. Thus, the PDF is calculated as $f_E(\theta) = \frac{1}{2}\sin\theta, \theta \in [0, \pi]$.□

As illustrated in Fig. 2, $\mathcal{S}_{I,k}$ is defined as $\mathcal{S}_{I,k} = \{n | n \in \mathcal{S}_k, R_k \le r_n \le r_{\text{vis},k}\}$ with $N_{I,k}$ elements, where $r_{\text{vis},k} =$

[3]As off-axis angle β denotes the angular separation between co-channel ground cells as observed from the satellite, the minimum off-axis angle $\beta_{\min}$ is given by $\sqrt{12}\phi_b$ when co-channel reuse ratio is set to 4 [30].

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$\sqrt{R_k^2+2R_ER_k}$ is the maximum distance from a visible satellite on the k-th spherical surface to receivers. Here, condition $R_k \le r_n \le r_{\text{vis},k}$ indicates that satellite n lies within the receiver's horizon. Therefore, the distribution of the interfering satellite number $N_{I,k}$ is described in the following corollary.

Corollary 1: Since the positions of all satellites are distributed independently, $N_{I,k}$ follows a binomial distribution, $N_{I,k} \sim \mathbb{B}(N_k, p_{I,k})$ with success probability $p_{I,k} = \frac{R_k}{2(R_E+R_k)}$, where $p_{I,k}$ represents the probability that a satellite on the k-th spherical surface lies within the receiver's horizon.

Proof: From basic geometry and Lemma 1, on k-th spherical surface, the visibility probability of interfering satellites is calculated as $p_{I,k} = \mathbb{P}\left(\cos\theta \ge \frac{R_E}{R_E+R_k}\right) = F_E\left(\arccos\left(\frac{R_E}{R_E+R_k}\right)\right) = \frac{R_k}{2(R_E+R_k)}$. □

It is worth noting that when the transmitting satellite s_m is located on the k-th spherical surface, the number of interfering satellites should be $\max\{N_{I,k}-1, 0\}$. Since there are numerous visible interference satellites in ultra-dense LEO constellations, this factor has a negligible impact on the aggregate interference power. Similar to the channel modeling framework in [32], we adopt independent Rician fading for all interference links, i.e., $h_n \sim \text{Rician}(\chi_R, \sigma_R), \forall n \in \mathcal{S}_I$. As noted in [33], we can approximate the aggregate interference power distribution using a Gamma distribution and estimate its parameters by the method of moments.

Theorem 1: The aggregate interference power I_{ag} approximately follows a Gamma distribution $\text{Gamma}(k_I, \theta_I)$ with PDF $f_I(x)$, a shape parameter $k_I = \frac{\mu_I^2}{\nu_I}N$, and a scale parameter $\theta_I = \frac{\nu_I}{\mu_I}$, where the mean $\bar{I}_{\text{ag}}$ and the variance σ_I^2 are calculated as $\bar{I}_{\text{ag}} = \mu_I N$ and $\sigma_I^2 = \nu_I N$ respectively, and μ_I and ν_I can be obtained through numerical methods.

Proof: The proof is provided in Appendix VII-A. □

According to Theorem 1, the aggregate interference power I_{ag} approximately follows a Gamma distribution, with both the mean $\bar{I}_{\text{ag}}$ and variance σ_I^2 varying linearly with satellite number N. The accuracy of this approximation is further validated through numerical simulations in Section VI-B.

IV. Fixed Transmitting Satellite: Covert Capacity Analysis With Interference Uncertainty

In this section, neglecting directional uncertainty temporarily, we analyze the impact of aggregate interference on the covert capacity, presenting how the capacity varies with slot length L and the number of satellites N. Considering a covert SatCom scenario involving a single accessible satellite and a single time slot, the set $\mathcal{S}_A$ is constrained to contain exactly one accessible satellite, making the transmitting satellite fixed. Consequently, Willie can focus its monitoring efforts exclusively on the predetermined transmitting satellite. Therefore, the receiving signals $\boldsymbol{y}_1, \boldsymbol{z}_1$ for Bob and Willie in a single slot are shown as (1) and (3) with setting $M=1$ and $\delta_{s_W,s_1}=1$.

From the analysis in Section II-B, Willie's receiving signal under hypotheses $\mathbf{H_0}$ and $\mathbf{H_1}$ is

$$\boldsymbol{z}_1 = \begin{cases} \boldsymbol{v}_{W,1} + \boldsymbol{i}_{W,1}, & \mathbf{H_0}, \\ \sqrt{P_{W,0}}\boldsymbol{c}_1 + \boldsymbol{v}_{W,1} + \boldsymbol{i}_{W,1}, & \mathbf{H_1}. \end{cases} \tag{12}$$

Subsequently, the test statistic is denoted as $T_W(1,L) = \frac{\boldsymbol{z}_1^H\boldsymbol{z}_1}{L}$, which is shown as $T_W(L)$ in this section for brevity. Then, the minimum error detection probability $\eta_{\min}(1,L)$ is concisely expressed as $\eta_{\min}(L)$ and calculated as follows,

$$\eta_{\min}(L) = \min_{\gamma}\{\mathbb{P}(T_W(L) > \gamma|\mathbf{H_0}) + \mathbb{P}(T_W(L) < \gamma|\mathbf{H_1})\}. \tag{13}$$

A. Detection Probability Analysis

According to (12), Willie's received signal can be shown as $\boldsymbol{z}_1 = (z_1[1], z_1[2], \cdots, z_1[L])$, where $z_1[l] \sim \mathcal{CN}(0, P_{W,R})$,

$$P_{W,R} = \begin{cases} I_{W,1} + \sigma_W^2, & \mathbf{H_0}, \\ I_{W,1} + \sigma_W^2 + P_{W,0}, & \mathbf{H_1}, \end{cases} \tag{14}$$

where the interference power $I_{W,1}$ is modeled as a random variable I_{ag} investigated in Section III, and its variation across different time slots affects Willie's signal detection. Based on the received signals $\boldsymbol{z}_1$, the test statistic is calculated,

$$T_W(L) = \frac{\boldsymbol{z}_1^H\boldsymbol{z}_1}{L} = \frac{1}{L}\sum_{l=1}^{L}|z_1[l]|^2 = \frac{P_{W,R}}{L}\sum_{l=1}^{L}|X_l|^2, \tag{15}$$

where $X_1, X_2, \cdots, X_L$ are independent and identically distributed (i.i.d.) standard complex Gaussian random variables. As the sum of squares of L i.i.d. standard complex Gaussian random variables follows a Gamma distribution $\text{Gamma}(L, 1/L)$, we can express the test statistic $T_W(L)$ as

$$T_W(L) = P_{W,R}V = \begin{cases} (I_{\text{ag}} + \sigma_W^2)V, & \mathbf{H_0}, \\ (I_{\text{ag}} + \sigma_W^2 + P_{W,0})V, & \mathbf{H_1}, \end{cases} \tag{16}$$

where $I_{W,1}$ is modeled as a random variable I_{ag} and $V \sim \text{Gamma}(L, 1/L)$. Therefore, substituting $T_W(L)$ into (13),

$$\begin{aligned} &\eta_{\min}(L) \\ &= \min_{\gamma}\{\mathbb{P}((I_{\text{ag}} + \sigma_W^2)V > \gamma) \\ &\quad + \mathbb{P}((I_{\text{ag}} + \sigma_W^2 + P_{W,0})V < \gamma)\} \\ &= 1 - \max_{\gamma}\mathbb{P}\left(\frac{\gamma}{V} - \sigma_W^2 - P_{W,0} \le I_{\text{ag}} \le \frac{\gamma}{V} - \sigma_W^2\right) \\ &= 1 - \max_{\gamma}\int_0^{\infty}\int_{\frac{\gamma}{V}-\sigma_W^2-P_{W,0}}^{\frac{\gamma}{V}-\sigma_W^2} f_I(I_{\text{ag}})\mathrm{d}I_{\text{ag}} f_V(V)\mathrm{d}V, \end{aligned} \tag{17}$$

where $f_V(V)$ denotes PDF of V.

Considering that the minimum error detection probability $\eta_{\min}(L)$ for finite L in (17) requires the evaluation of double integrals, an approximation method is provided. As the signal $\boldsymbol{c}_1$ is statistically independent of noise $\boldsymbol{v}_{W,1}$ and interference $\boldsymbol{i}_{W,1}$, the test statistic under hypothesis $\mathbf{H}_1$ is approximated,

$$\begin{aligned} T_W(L) &\approx P_{W,0}\frac{\boldsymbol{c}_1^H\boldsymbol{c}_1}{L} + \frac{(\boldsymbol{v}_{W,1}+\boldsymbol{i}_{W,1})^H(\boldsymbol{v}_{W,1}+\boldsymbol{i}_{W,1})}{L} \\ &\overset{(a)}{=} P_{W,0} + (I_{\text{ag}} + \sigma_W^2)V, \end{aligned} \tag{18}$$

where (a) is obtained due to the fact that the transmitting satellite normalizes the energy of the transmitted signal, i.e., $\boldsymbol{c}_1^H\boldsymbol{c}_1 = L$. Substituting $T_W(L)$ under $\mathbf{H}_0$ and $\mathbf{H}_1$ into (13), $\eta_{\min}(L)$ is approximated as

$$\eta_{\min}(L) \approx 1 - \max_{\gamma}\mathbb{P}(\gamma - P_{W,0} \le P_{\text{IN}} \le \gamma), \tag{19}$$

where $P_{\text{IN}} = (I_{\text{ag}} + \sigma_W^2)V$ denotes the average power of interference and noise, and is a random variable.

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Based on the approximation of $I_{\rm ag}$ in Theorem 1, we derive a lower bound for the minimum error detection probability under finite slot lengths. Furthermore, by considering a sufficiently large number of satellites inherent to ultra-dense LEO constellations, a more concise lower bound can be established.

Lemma 2: The minimum error detection probability $\eta_{\min}(L)$ is bounded as follows,

$$\eta_{\min}(L) \geq \begin{cases} \dfrac{\Gamma(k_T, P_{W,0}/\theta_T)}{\Gamma(k_T)}, & 0 \leq k_T \leq 1, \\ 1 - \dfrac{P_{W,0}(k_T-1)^{k_T-1}}{\theta_T e^{k_T-1}\Gamma(k_T)}, & k_T > 1, \end{cases} \tag{20}$$

and with sufficiently large N and L,

$$\eta_{\min}(L) \geq 1 - \frac{P_{W,0}}{\sqrt{2\pi k_T \theta_T^2}}, \tag{21}$$

where $\Gamma(k_T, x)$ represents the upper incomplete gamma function, $\Gamma(k_T)$ represents the gamma function, and $k_T\theta_T = \mu_I N$ and $k_T\theta_T^2 = \nu_I N + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{L}$.

Proof: The proof is provided in Appendix VII-B. □

From (20) in Lemma 2, the detection error probability decreases with $P_{W,0}$, consistent with the intuition that stronger signals are easier to detect. Under sufficiently large N and L, the simplified form in (21) reveals that $\eta_{\min}(L)$ increases monotonically with the interference power mean $\mu_I N$, variance $\nu_I N$, and noise power σ_W^2, indicating that both noise and interference degrade Willie's detection capability. Moreover, increasing L reduces the variance of $P_{\rm IN}$, mitigates uncertainty, and thereby lowers $\eta_{\min}(L)$.

As L approaches infinity, the minimum error detection probability is bounded as $\eta_{\min}(\infty) = \lim_{L\to\infty}\eta_{\min}(L) \geq 1 - \frac{P_{W,0}}{\sqrt{2\pi\nu_I N}}$, which is determined by $P_{W,0}$ and the variance $\nu_I N$ of interference $I_{\rm ag}$. Next, the relationship between the $\eta_{\min}(L)$ and $\eta_{\min}(\infty)$ is described in the following Lemma.

Lemma 3: Willie's minimum error detection probability $\eta_{\min}(L)$ with slot length L is no less than Willie's minimum error detection probability $\eta_{\min}(\infty)$ with the infinite slot length $L \to \infty$, i.e, $\eta_{\min}(L) \geq \eta_{\min}(\infty)$.

Proof: As the frame length approaches infinity, the average energy $T_W(\infty) = \lim_{L\to\infty}\frac{1}{L}\sum_{l=1}^{L}|z[l]|^2 = P_{W,R}$, where $P_{W,R}$ is defined in (14). Therefore, minimum error detection probability $\eta_{\min}(\infty)$ with $L \to \infty$ is calculated as

$$\begin{aligned} &\eta_{\min}(\infty) \\ &= \min_{\gamma}\{\mathbb{P}(I_{\rm ag} + \sigma_W^2 > \gamma) + \mathbb{P}(I_{\rm ag} + \sigma_W^2 + P_{W,0} < \gamma)\} \\ &= 1 - \max_{\gamma}\mathbb{P}(\gamma - \sigma_W^2 - P_{W,0} \leq I_{\rm ag} \leq \gamma - \sigma_W^2) \\ &= 1 - \max_{\gamma}\int_{\gamma-\sigma_W^2-P_{W,0}}^{\gamma-\sigma_W^2} f_I(I_{\rm ag})\mathrm{d}I_{\rm ag}, \end{aligned} \tag{22}$$

□

According to (17), $\eta_{\min}(L)$ can be calculated as

$$\begin{aligned} &\eta_{\min}(L) \\ &\geq 1 - \int_0^{\infty}\max_{\gamma}\int_{\frac{\gamma}{V}-\sigma_W^2-P_{W,0}}^{\frac{\gamma}{V}-\sigma_W^2} f_I(I_{\rm ag})\mathrm{d}I_{\rm ag} f_V(V)\mathrm{d}V \\ &= 1 - \max_{\gamma}\int_{\gamma-\sigma_W^2-P_{W,0}}^{\gamma-\sigma_W^2} f_I(I_{\rm ag})\mathrm{d}I_{\rm ag} \\ &= \eta_{\min}(\infty). \end{aligned} \tag{23}$$

From Lemma 3, the minimum error detection probability $\eta_{\min}(L)$ is no less than $\eta_{\min}(\infty)$, revealing the degradation in Willie's detection capability when $L \to \infty$.

B. Outage Capacity Analysis

For legitimate user Bob, considering that the interference power $I_{B,1}$ is modeled as a random variable $I_{\rm ag}$, the outage probability $p_{\rm out}$ at Bob under communication rate R can be calculated as follows,

$$\begin{aligned} p_{\rm out}(P_{B,0}, R) &= \mathbb{P}\left(\log\left(1 + \frac{P_{B,0}}{\sigma_B^2 + I_{B,1}}\right) \leq R\right) \\ &= \mathbb{P}\left(I_{\rm ag} \geq \frac{P_{B,0}}{2^R - 1} - \sigma_B^2\right), \end{aligned} \tag{24}$$

where $p_{\rm out}(P_{B,0}, R) = \mathbb{P}(I_{\rm ag} \geq I_{\rm out})$ and $R = \log\left(\frac{P_{B,0}}{\sigma_B^2 + I_{\rm out}}\right)$ by defining $I_{\rm out} = \frac{P_{B,0}}{2^R-1} - \sigma_B^2$.

To ensure the covertness of the signal in the covert SatCom system, the minimum error detection probability $\eta_{\min}(L)$ by Willie is limited by $1-\epsilon$. Additionally, a maximum outage probability κ is imposed to ensure reliability. In summary, the maximum communication rate, subject to the constraints of minimum error detection probability $\eta_{\min}(L) \geq 1-\epsilon$ and outage probability $p_{\rm out} \leq \kappa$, is defined as the covert capacity $C_{\rm cov}(L)$ for the covert SatCom system with slot length L,

$$\begin{aligned} C_{\rm cov}(L) = &\max_{P_{B,0}, I_{\rm out}} \log\left(\frac{P_{B,0}}{\sigma_B^2 + I_{\rm out}}\right), \\ &s.t.\ \mathrm{C1}: \eta_{\min}(L) \geq 1-\epsilon, \\ &\mathrm{C2}: \mathbb{P}(I_{\rm ag} \geq I_{\rm out}) \leq \kappa. \end{aligned} \tag{25}$$

According to (19), the covert capacity $C_{\rm cov}(L)$ can be transformed into

$$\begin{aligned} C_{\rm cov}(L) \overset{(a)}{\approx} &\max_{P_{W,0}, I_{\rm out}} \frac{G_B P_{W,0}}{G_W(\sigma_B^2 + I_{\rm out})}, \\ &s.t.\ \mathrm{C3}: \max_{\gamma}\mathbb{P}(\gamma - P_{W,0} \leq P_{\rm IN} \leq \gamma) \leq \epsilon, \\ &\mathrm{C2}: \mathbb{P}(I_{\rm ag} \geq I_{\rm out}) \leq \kappa, \end{aligned} \tag{26}$$

where (a) is obtained as that $\log\left(1 + \frac{P_{B,0}}{\sigma_B^2 + I_{\rm out}}\right) \approx \frac{P_{B,0}}{\sigma_B^2 + I_{\rm out}}$ and $P_{B,0} = \frac{G_B P_{W,0}}{G_W}$, and (C3) is obtained as $\eta_{\min}(L) \approx 1 - \max_{\gamma}\mathbb{P}(\gamma - P_{W,0} \leq P_{\rm IN} \leq \gamma) \geq 1-\epsilon$.

Theorem 2: The covert capacity $C_{\rm cov}(L)$ with finite slot length L can be approximated as follows,

$$C_{\rm cov}(L) \approx \begin{cases} \dfrac{G_B\theta_T\Gamma^{-1}(k_T, (1-\epsilon)\Gamma(k_T))}{G_W\left(\sigma_B^2 + \theta_I\Gamma^{-1}(k_I, \kappa\Gamma(k_I))\right)}, & 0 \leq k_T \leq 1, \\ \dfrac{G_B\epsilon\frac{\theta_T e^{k_T-1}\Gamma(k_T)}{(k_T-1)^{k_T-1}}}{G_W\left(\sigma_B^2 + \theta_I\Gamma^{-1}(k_I, \kappa\Gamma(k_I))\right)}, & k_T > 1, \end{cases} \tag{27}$$

and with sufficiently large N and L,

$$C_{\rm cov}(L) \approx \frac{\sqrt{2\pi}G_B\epsilon\sqrt{\nu_I N + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{L}}}{G_W(\sigma_B^2 + \mu_I N + \sqrt{\nu_I N}Q^{-1}(\kappa))}, \tag{28}$$

where $\Gamma^{-1}(k_I, x)$ is the inverse function of $\Gamma(k_I, x)$

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Proof: According to (20) in Lemma 2, $P_{W,0}$ is calculated as

$$P_{W,0} \leq \begin{cases} \theta_T \Gamma^{-1}(k_T, (1-\epsilon)\Gamma(k_T)), & 0 \leq k_T \leq 1, \\ \dfrac{\theta_T e^{k_T-1}\Gamma(k_T)\epsilon}{(k_T-1)^{k_T-1}}, & k_T > 1, \end{cases} \tag{29}$$

Based on Theorem 1, (C2) is shown as follows,

$$\mathbb{P}(I_{\text{ag}} \geq I_{\text{out}}) = \frac{\Gamma(k_I, I_{\text{out}}/\theta_T)}{\Gamma(k_I)} \leq \kappa \tag{30}$$

Therefore, $I_{\text{out}} \geq \theta_I \Gamma^{-1}(k_I, \kappa\Gamma(k_I))$. Substituting corresponding $P_{W,0}$ and I_{out} into (26), $C_{\text{cov}}(L)$ in (27) is obtained.

With sufficiently large N and L, $P_{W,0}$ is calculated according to (21) in Lemma 2 as

$$P_{W,0} \leq \sqrt{2\pi}\epsilon\sqrt{\nu_I N + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{L}}. \tag{31}$$

By approximating the Gamma random variable with a Gaussian variable under sufficiently large $k_I = \frac{\mu_I^2}{\nu_I}N$, $\mathbb{P}(I_{\text{ag}} \geq I_{\text{out}}) \approx Q\left(\frac{I_{\text{out}}-\bar{I}_{\text{ag}}}{\sigma_I}\right) \leq \kappa$. Therefore, $I_{\text{out}} \geq \bar{I}_{\text{ag}} + \sigma_I Q^{-1}(\kappa) = \mu_I N + \sqrt{\nu_I N}Q^{-1}(\kappa)$.

In summary, substituting corresponding $P_{W,0}$ and I_{out} into (26), $C_{\text{cov}}(L)$ in (28) is obtained. □

From Theorem 2, the covert capacity $C_{\text{cov}}(L)$ gradually decreases with the slot length L, and increases with ϵ. The covert capacity also increases with the outage probability κ, though the effect is relatively minor. This observation is further validated through numerical simulations in Section VI. When N is relatively small, the interference power $\bar{I}_{\text{ag}} = \mu_I N$ remains lower than the fixed noise power σ_W^2, thus $C_{\text{cov}}(L)$ is primarily influenced by the numerator and increase with N. However, as N further increases, $\mu_I N$ exceeds σ_W^2, which diminishes the growth of $C_{\text{cov}}(L)$ and may even lead to its decrease.

Moreover, $\sqrt{\nu_I N}$ and $\mu_I N$ are typically comparable in practice. Based on this, for small κ and large N, L, the denominator in (28) is dominated by $\mu_I N$, while the numerator scales with $\nu_I N$. Consequently, the variance of interference power benefits covert communication while its mean is detrimental, consistent with the intuition that uncertainty of interference power weakens the warden's detection, whereas its mean reduces Bob's received SNR.

Remark 1: When the slot length $L \to \infty$, the covert capacity approaches a non-zero limit, expressed as

$$C_{\text{cov}}(\infty) \approx \frac{G_B \epsilon\sqrt{2\pi\nu_I N}}{G_W(\sigma_B^2 + \mu_I N + \sqrt{\nu_I N}Q^{-1}(\kappa))}. \tag{32}$$

In conventional covert communication over an AWGN channel, the covert capacity is related to the energy uncertainty of Gaussian noise. Nevertheless, as the slot length L increases, the estimated noise power gradually converges to its variance σ_W^2. Therefore, as $L \to \infty$, the warden can reliably detect any signal with non-zero received power P_W by setting the detection threshold to $\sigma_W^2 + P_W/2$, which results in a zero asymptotic covert capacity.

However, in ultra-dense LEO constellations with aggregate interference, the covert capacity depends on the total energy uncertainty of both channel noise and aggregate interference. Due to the uncertainty of aggregate interference power I_{ag}, the estimation of this total energy does not readily converge as L increases. Consequently, when $L \to \infty$, Willie cannot reliably distinguish between a signal consisting solely of interference and noise and a signal containing information using any fixed power threshold, thereby allowing the transmitter to achieve a positive covert capacity with sufficiently low transmit power. In particular, the uncertainty of I_{ag} is roughly quantified by its standard deviation $\sigma_I = \sqrt{\nu_I N}$. As revealed in (32), this standard deviation explicitly appears in the numerator of the covert capacity, confirming that the non-zero variance of aggregate interference directly enables the achievement of a positive covert capacity.

V. Multiple Accessible Satellites: Covert Capacity Analysis With Directional Uncertainty

Building upon the single-satellite covert SatCom scheme discussed in Section IV, we further analyze the covert capacity for the multi-satellite covert SatCom schemes leveraging both interference power uncertainty and directional uncertainty.

A. Detection Probability Analysis

As analyzed in Section II, one target accessible satellite s_m will be randomly selected from $\mathcal{S}_A$ for transmitting covert signals to Bob, while Willie monitors a fixed satellite s_W among $\mathcal{S}_A$ across all M slots, collecting the data set $\{\boldsymbol{z}_m\}_{m=1}^M$. However, prior to signal detection, it is impossible to determine which of the monitored data contains information. Therefore, all these data must undergo detection, and the test statistic $T_W(M, L)$ is calculated according to (8).

According to (7), Willie's received signals are shown as $\boldsymbol{z}_m = \boldsymbol{v}_{W,m} + \boldsymbol{i}_{W,m}$ under hypothesis $\mathbf{H}_0$. As $z_m[l] \sim \mathcal{CN}(0, I_{W,m} + \sigma_W^2)$ for $m \in \mathcal{M}$ and $1 \leq l \leq L$, the test statistic under hypothesis $\mathbf{H}_0$ is calculated as

$$T_W(M, L) = \frac{1}{M}\sum_{m=1}^{M}(I_{W,m} + \sigma_W^2)V_m \triangleq \frac{1}{M}\sum_{m=1}^{M} P_{\text{IN},m}. \tag{33}$$

where $P_{\text{IN},m} = (I_{\text{ag},m} + \sigma_W^2)V_m, V_m \sim \text{Gamma}(L, 1/L)$ for $m \in \mathcal{M}$, since $I_{W,m}$ is modeled as a random variable I_{ag} with PDF $f_I(x)$ in Section III, and $I_{\text{ag},m}$ for each m is modeled as an i.i.d. realization of I_{ag}.

Similarly, as $\frac{\boldsymbol{z}_m^H \boldsymbol{z}_m}{L} \approx \delta_{s_W,s_m}P_{W,0} + P_{\text{IN},m}$ under hypothesis $\mathbf{H}_1$, corresponding $T_W(M, L)$ is calculated as follows,

$$\begin{aligned} T_W(M, L) &= \frac{1}{M}\sum_{m=1}^{M}(\delta_{s_W,s_m}P_{W,0} + P_{\text{IN},m}) \\ &= B\frac{P_{W,0}}{M} + \frac{1}{M}\sum_{m=1}^{M} P_{\text{IN},m}, \end{aligned} \tag{34}$$

where $B \sim \text{Binomial}\left(M, \frac{1}{N_A}\right)$ as δ_{s_W,s_m} has a $\frac{1}{N_A}$ probability of being equal to 1. Therefore, substituting (33) and (34) into (9), $\eta_{\min}(M, L)$ is obtained as follows,

$$\eta_{\min}(M, L) \approx 1 - \max_{\gamma}\mathbb{P}\left(\gamma - B\frac{P_{W,0}}{M} \leq P_{\text{IN,avg}} \leq \gamma\right), \tag{35}$$

where $P_{\text{IN,avg}} = \frac{1}{M}\sum_{m=1}^{M} P_{\text{IN},m}$ is a random variable.

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Based on the approximation of I_{ag} in Theorem 1, a concise lower bound is established with sufficiently large N and L.

Lemma 4: When the number of satellites N and slot length L are sufficiently large, the minimum error detection probability $\eta_{\min}(M,L)$ can be bounded as follows,

$$\eta_{\min}(M,L) \geq 1 - \frac{P_{W,0}}{N_A\sqrt{2\pi k_T^{(M)}(\theta_T^{(M)})^2}}, \tag{36}$$

where $k_T^{(M)}\theta_T^{(M)} = \mu_I N$ and $k_T^{(M)}(\theta_T^{(M)})^2 = \frac{\nu_I N}{M} + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{ML}$.

Proof: The proof is provided in Appendix VII-C. □

Similar to the single-satellite case, $\eta_{\min}(M,L)$ is monotonically increasing in the interference power mean $\mu_I N$, variance $\nu_I N$, and noise power σ_W^2, and decreasing in L. It also decreases with the number of slots M, since more slots provide Willie with additional observations that enhance his detection capability. Moreover, $\eta_{\min}(M,L)$ increases with the number of accessible satellites N_A, reflecting the adverse effect of directional uncertainty on Willie's detection.

B. Outage Capacity Analysis

In the multi-satellite cooperative covert scheme, Bob's average communication rate over M slots was calculated as

$$C_{\text{avg}} = \frac{1}{M}\log\left(1 + \frac{MP_{B,0}}{\sigma_B^2 + \frac{1}{M}\sum_{m=1}^{M} I_{\text{ag},m}}\right), \tag{37}$$

where the signals of M slots are coherent, and the average interference power for M slots is calculated as $I_{\text{avg}} = \frac{1}{M}\sum_{m=1}^{M} I_{\text{ag},m}$. Note that Willie cannot acquire the phase information prior to detection. The outage probability of M slots at Bob under communication rate R can be calculated,

$$\begin{aligned} &p_{\text{out}}(P_{B,0},R) \\ &= \mathbb{P}\left(\frac{1}{M}\log\left(1 + \frac{MP_{B,0}}{\sigma_B^2 + \frac{1}{M}\sum_{m=1}^{M} I_{\text{ag},m}}\right) \leq R\right) \\ &= \mathbb{P}(I_{\text{avg}} \geq I_{\text{out}}), \end{aligned} \tag{38}$$

where $I_{\text{out}} = \frac{MP_{B,0}}{2^{MR}-1} - \sigma_B^2$. According to Theorem 1, I_{avg} is the sum of M i.i.d. Gamma random variables, and it also satisfies a Gamma distribution, i.e., $I_{\text{avg}} \sim \text{Gamma}(Mk_I, \theta_I/M)$.

Therefore, similarly to Section IV-B, under the constraint of minimum error detection probability $\eta_{\min}(M,L) \geq 1-\epsilon$ and the outage probability constraint $p_{\text{out}} \leq \kappa$, the covert capacity $C_{\text{cov}}(M,L)$ is calculated as follows,

$$\begin{aligned} &C_{\text{cov}}(M,L) \\ &\overset{(a)}{\approx} \max_{P_{W,0}, I_{\text{out}}} \frac{G_B P_{W,0}}{G_W(\sigma_B^2 + I_{\text{out}})}, \\ &\quad s.t.\ \text{C4}: \max_{\gamma} \mathbb{P}\left(\gamma - B\frac{P_{W,0}}{M} \leq P_{\text{IN,avg}} \leq \gamma\right) \leq \epsilon, \\ &\qquad \text{C5}: \mathbb{P}(I_{\text{avg}} \geq I_{\text{out}}) \leq \kappa, \end{aligned} \tag{39}$$

where (a) is because $R = \frac{1}{M}\log\left(1 + \frac{MP_{B,0}}{\sigma_B^2 + I_{\text{out}}}\right) \approx \frac{P_{B,0}}{\sigma_B^2 + I_{\text{out}}}$ and $P_{B,0} = \frac{G_B P_{W,0}}{G_W}$, and (C4) is obtained according to (35).

Theorem 3: With sufficiently large satellite number N and slot length L, the covert capacity $C_{\text{cov}}(M,L)$ is approximated,

$$C_{\text{cov}}(M,L) \approx \frac{\sqrt{2\pi}N_A\epsilon G_B\sqrt{\frac{\nu_I N}{M} + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{ML}}}{G_W\left(\sigma_B^2 + \mu_I N + \sqrt{\frac{\nu_I N}{M}}Q^{-1}(\kappa)\right)}. \tag{40}$$

Proof: When the satellite number N and slot length L are sufficiently large, the power of received signal $P_{W,0}$ is calculated according to Lemma 4, that is

$$P_{W,0} \leq \sqrt{2\pi}N_A\epsilon\sqrt{\frac{\nu_I N}{M} + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{ML}}. \tag{41}$$

Given that $k_I = \frac{\mu_I^2}{\nu_I}N$, the Gamma distribution can be approximated by a Gaussian distribution for sufficiently large N. As $I_{\text{avg}} \sim \text{Gamma}(Mk_I, \theta_I/M)$, (C5) is shown as

$$\mathbb{P}(I_{\text{avg}} \geq I_{\text{out}}) \approx Q\left(\frac{I_{\text{out}} - k_I\theta_I}{\sqrt{k_I\theta_I^2/M}}\right) \leq \kappa \tag{42}$$

Therefore, leveraging Theorem 1,

$$I_{\text{out}} \geq \mu_I N + \sqrt{\frac{\nu_I N}{M}}Q^{-1}(\kappa). \tag{43}$$

By substituting (41) and (43) into (39), an approximate expression for $C_{\text{cov}}(M,L)$ is obtained. □

From Theorem 3, $C_{\text{cov}}(M,L)$ increases with κ and ϵ, but decreases with larger L and M. Considering the number of accessible satellites $N_A = \rho_A N$, $C_{\text{cov}}(M,L)$ increases monotonically with the number of satellites N. Based on basic geometric relationships, N_A can be expressed in detail as

$$N_A \approx \left\lfloor \frac{N}{2K}\sum_{k=1}^{K}\left(1 - \sin\left(\zeta_{\min} + \arcsin\frac{R_E\cos\zeta_{\min}}{R_E + R_k}\right)\right)\right\rfloor, \tag{44}$$

and $\zeta_{\min}$ denotes the minimum elevation angle of the accessible satellites. It is straightforward to verify that $\frac{dN_A}{d\zeta_{\min}}$ remains strictly negative over the elevation angle range $[0, \pi/2]$, which implies that N_A decreases monotonically with $\zeta_{\min}$. According to (40), $C_{\text{cov}}(M,L)$ also decreases monotonically with $\zeta_{\min}$.[4]

Remark 2: As slot length L approaches infinity, the covert capacity is approximated as follows,

$$C_{\text{cov}}(M,\infty) \approx \frac{\sqrt{2\pi}N_A\epsilon G_B\sqrt{\nu_I N}}{G_W\left(\sqrt{M}(\sigma_B^2 + \mu_I N) + \sqrt{\nu_I N}Q^{-1}(\kappa)\right)}. \tag{45}$$

Compared with the single-satellite case, the improvement in covert capacity with multiple satellites is mainly determined by the number of accessible satellites $N_A = \rho_A N$, indicating that the covert capacity will continue to grow with the ongoing expansion of LEO constellations.

Remark 3: With D beams, Willie can monitor more satellites simultaneously, collecting up to MD frames over M

[4]For smaller operators or early-stage deployments with a limited number of satellites, a relatively small N_A may degrade the covert capacity. This limitation can be mitigated by lowering the minimum elevation angle $\zeta_{\min}$, thereby increasing N_A, as in the early Starlink design with a smaller $\zeta_{\min}$ [26].

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slots, where $D \leq N_A$ in general. Accordingly, the detection statistic $T_W(M, L, D)$ under $\mathbf{H}_1$ is modified as follows,

$$T_W(M, L, D) = B' \frac{P_{W,0}}{MD} + \frac{\sum_{d=1}^{D} \sum_{m=1}^{M} P_{\mathrm{IN},m,d}}{MD}, \quad (46)$$

where $B' \sim \mathrm{Binomial}(M, D/N_A)$ denotes the number of successfully intercepted frames carrying information over M slots, and $P_{\mathrm{IN},m,d} = (I_{W,m,d} + \sigma_W^2)V_{m,d}, V_{m,d} \sim \mathrm{Gamma}(L, 1/L)$. Similarly, $T_W(M, L, D)$ under $\mathbf{H}_0$ can be obtained by removing the signal component $B' \frac{P_{W,0}}{MD}$ from (46). Since the eavesdropping mechanism does not affect legitimate reception, the covert capacity can be derived as follows,

$$C_{\mathrm{cov}}(M, L, D) \approx \frac{\sqrt{2\pi} N_A \epsilon G_B \sqrt{\frac{\nu_I N}{M} + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{ML}}}{G_W \sqrt{D} \left(\sigma_B^2 + \mu_I N + \sqrt{\frac{\nu_I N}{M}} Q^{-1}(\kappa) \right)}. \quad (47)$$

From (47), when Willie is equipped with D beams, the covert capacity is reduced by a factor of $1/\sqrt{D}$ compared with the result in Theorem 3. This results from the observation that Willie can intercept D times more frames, thereby reducing the standard deviation of the averaged interference-plus-noise energy in (46) to $1/\sqrt{D}$ of that in (34), which enhances his detection capability and consequently reduces the system's covert capacity. Moreover, the number of beams D cannot increase indefinitely, as it is constrained by practical system complexity and hardware cost [34].

C. Outage Capacity Analysis With Oracle Information

Furthermore, in order to enhance Willie's detection performance for comparison, we assume the existence of an oracle that reveals to Willie which slots he successfully captures, allowing him to exclude erroneous slots $\{\boldsymbol{z}_m | \delta_{s_W, s_m} = 0\}$ from the statistic. In practice, this oracle information can be interpreted as an extreme benchmark, which represents the best possible performance achievable if perfect knowledge of successfully intercepted frames. Specifically, if Willie were equipped with unlimited computational resources or sufficient observation time, he could theoretically attempt all possible combinations of received frame sequences. The optimal outcome of such exhaustive search would be to coherently combine the energies of all correctly captured frames while discarding the erroneous ones. By assuming the availability of oracle information, we establish this upper bound on Willie's detection capability, which serves as a useful reference for evaluating the robustness of the proposed covert communication scheme under adversarial conditions.

Therefore, the subset of received signals containing the attenuated covert signal can be expressed as $\mathcal{M}_C = \{m | \delta_{s_W, s_m} = 1, m \in \mathcal{M}\}$. Leveraging this oracle information, Willie's test statistic becomes

$$T_W^{\mathrm{ora}}(M, L) = \frac{1}{M_C} \sum_{m \in \mathcal{M}_C} \frac{\boldsymbol{z}_m^H \boldsymbol{z}_m}{L} \underset{D_0}{\overset{D_1}{\gtrless}} \gamma, \quad (48)$$

where M_C represents the number of elements in the set $\mathcal{M}_C$. Therefore, the covert capacity under this assumption is presented in the following Theorem.

TABLE I
PARAMETERS SETTING OF LEO SATELLITES AND USERS

	Parameter	Setting
LEOs	Frequency	13.5 GHz
	Co-channel reuse factor	4
	Bandwidth	250 MHz
	Transmitting power P_s	15 dBW
	Transmitting antenna gain G_A	35 dBi
Users	Receiving antenna gain G_B, G_W	35 dBi
	System temperature	362 K
Interference links	Rician factor K	16 dB

Theorem 4: When the satellite number N and slot length L are sufficiently large, the covert capacity with oracle information, M slots, and N_A accessible satellites is approximated,

$$C_{\mathrm{cov}}^{\mathrm{ora}}(M, L) \approx \frac{\sqrt{2\pi} G_B \epsilon \sqrt{\nu_I N + \frac{\nu_I N + (\mu_I N + \sigma_W^2)^2}{L}}}{E\{\sqrt{M_C}\} G_W \left(\sigma_B^2 + \mu_I N + \sqrt{\frac{\nu_I N}{M}} Q^{-1}(\kappa) \right)}, \quad (49)$$

where $M_C \sim \mathrm{Binomial}\left(M, \frac{1}{N_A}\right)$.

Proof: The proof is provided in Appendix VII-D. □

As $M_C \sim \mathrm{Binomial}\left(M, \frac{1}{\rho_A N}\right)$, we find that $E\{\sqrt{M_C}\} \approx \sqrt{\frac{M}{\rho_A N}}$ when $\frac{1}{\rho_A N}$ is not particularly small. Consequently, the covert capacity $C_{\mathrm{cov}}^{\mathrm{ora}}(M, \infty)$ can be approximated as

$$C_{\mathrm{cov}}^{\mathrm{ora}}(M, \infty) \approx \frac{\sqrt{2\pi} G_B \epsilon \sqrt{\rho_A \nu_I}}{G_W \left(\sqrt{M} \left(\frac{\sigma_B^2}{N} + \mu_I \right) + \sqrt{\frac{\nu_I}{N}} Q^{-1}(\kappa) \right)}, \quad (50)$$

which increases slowly with N. However, when N continues to increase, $E\{\sqrt{M_C}\} \approx \mathbb{P}(M_C = 1) = \frac{M}{\rho_A N}$ as $\frac{1}{\rho_A N} \to 0$, the covert capacity can be approximately expressed as

$$C_{\mathrm{cov}}^{\mathrm{ora}}(M, \infty) \approx \frac{\sqrt{2\pi} G_B \rho_A \epsilon \sqrt{\nu_I N}}{G_W M \left(\frac{\sigma_B^2}{N} + \mu_I + \sqrt{\frac{\nu_I}{NM}} Q^{-1}(\kappa) \right)}, \quad (51)$$

which still increases with N.

VI. Numerical Simulation

In this section, we present numerical results to study the performance of proposed covert SatCom schemes and verify the theoretical findings.

A. Simulation Parameters

As of 27 March 2025, about 11,900 satellites are in low Earth orbit according to The Guardian [35]. In our simulations, the total number of satellites is set to $N = 12{,}000$, and the number of slots is $M = 100$. The LEO satellite parameters are chosen with reference to the Starlink configuration [36], [37]. Specifically, the LEO satellites are assumed to be uniformly distributed over two spherical shells at altitudes of 530 km and 550 km, which correspond to Starlink orbital heights. The satellite density is evenly divided across the two layers, with $\rho_1 = \rho_2 = 0.5$. The slot length is set to $L = 302$ [38], and the number of accessible satellites N_A is defined as the total number of satellites with elevation angles above 40° [36]. Additional system parameters are summarized in Table I, following [36], [37], [39].

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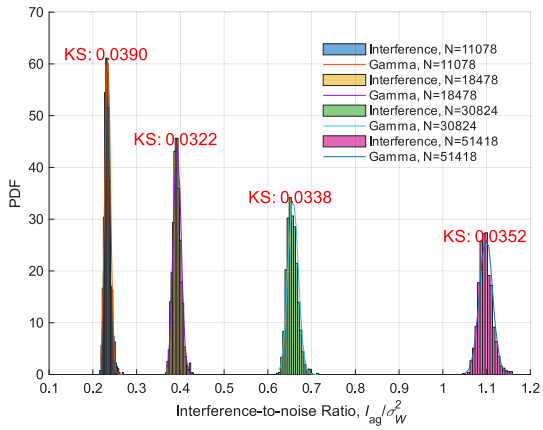

Fig. 3. Comparison and KS distances between the aggregate interference power and its Gamma approximation.

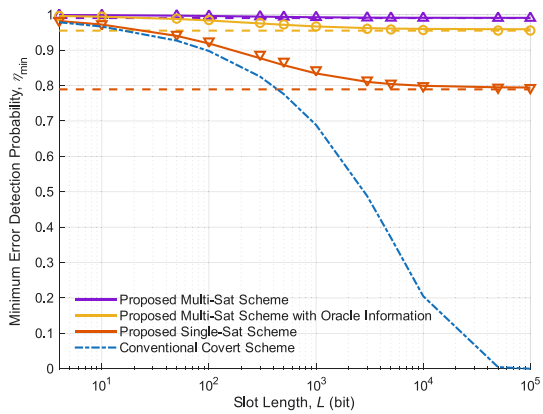

Fig. 4. Minimum error detection probability $\eta_{\min}$ versus slot length L: theoretical curves (solid lines), simulation results (dots), and asymptotic simulations for $L \to \infty$ (dashed lines).

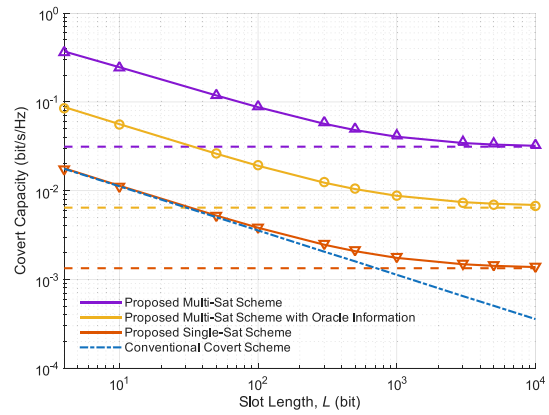

Fig. 5. Covert capacity versus slot length L under $\epsilon = 0.1$ and $\kappa = 0.01$: theoretical curves (solid lines), simulation results (dots), and asymptotic simulations for $L \to \infty$ (dashed lines).

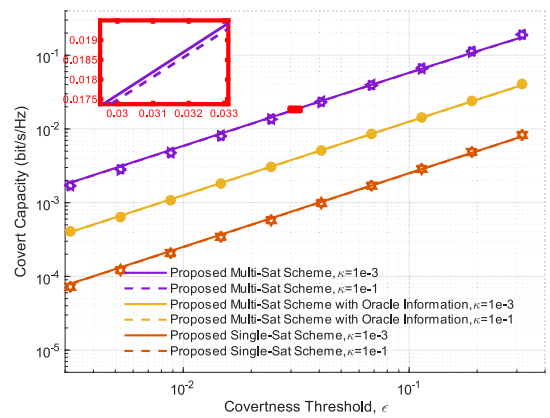

Fig. 6. Covert capacity versus covertness threshold ϵ under different outage probabilities $\kappa \in \{0.1, 0.001\}$: theoretical curves (lines) and simulation results (dots).

B. Aggregated Interference Distribution

Given that Willie's noise power is a fixed constant denoted by σ_W^2, it is used as a baseline for illustrating the relative strength of the interference power I_{ag}. To validate the approximation in Theorem 2, Fig. 3 provides simulation results of the interference-to-noise ratio I_{ag}/σ_W^2, along with the corresponding Gamma distribution. It can be observed that the Gamma approximation closely matches the simulated data distribution, as the Kolmogorov-Smirnov (KS) distances remain within an acceptable range.

C. Minimum Error Detection Probability and Covert Capacity

First, we analyze the effect of slot length L on Willie's minimum error detection probability $\eta_{\min}$ under a fixed received signal energy $P_{W,0} = -113$ dBm. As shown in Fig. 4, $\eta_{\min}$ of the traditional noise-assisted scheme decreases rapidly with L until it reaches 0, indicating that the wireless signals become highly detectable as L increases. In contrast, as shown by the simulation results, $\eta_{\min}$ of the proposed scheme also declines with L but approaches a nonzero limit as L approaches infinity, as illustrated by the horizontal dashed line in the figure. Furthermore, $\eta_{\min}$ of the multi-satellite (multi-Sat) scheme is significantly higher than that of the single-satellite (single-Sat) scheme, highlighting the detection challenges introduced by the transmitting satellite uncertainty. Even with oracle information, where Willie knows the set of successfully intercepted slots $\mathcal{M}_C$, $\eta_{\min}$ still remains higher than that of the single-Sat scheme, though lower than that of the multi-Sat scheme.

To comprehensively assess the impact of aggregate interference on both Willie's detection and Bob's communication, the covert capacity under $\epsilon = 0.1$ and $\kappa = 0.01$ is shown in Fig. 5. When L is small, the covert capacity of the noise-assisted covert satellite scheme is comparable to that of the proposed single-Sat scheme, as the noise energy within the limited number of bits still exhibits uncertainty. However, as L increases, the covert capacity of the traditional noise-assisted scheme declines following the square root law, scaling as $1/\sqrt{L}$. In contrast, the proposed scheme also declines with L initially but eventually converges to a nonzero asymptotic value as $L \to \infty$, as shown by the dashed line representing the asymptotic simulation results. Additionally, the multi-Sat scheme significantly outperforms the single-Sat scheme in covert capacity, while the multi-Sat scheme with oracle information achieves a performance level that lies between the two.

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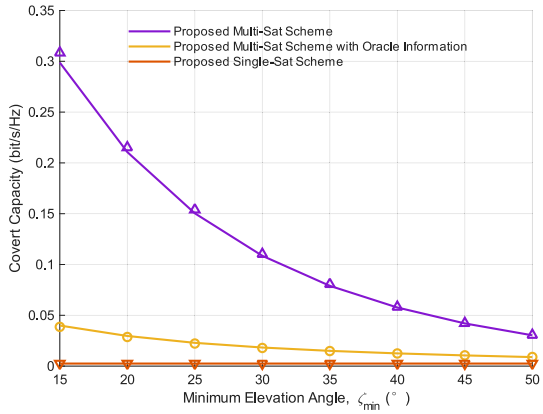

Fig. 7. Covert capacity versus minimum elevation angles $\zeta_{\min}$ under $L = 302$, $\epsilon = 0.1$, $\kappa = 0.01$: theoretical curves (lines) and simulation results (dots).

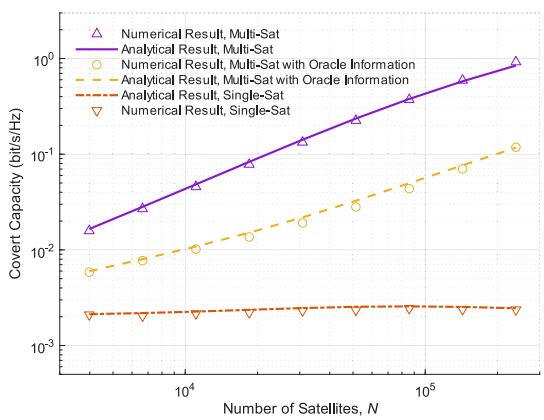

Fig. 8. Covert capacity versus number of satellites N under $L = 302$, $\epsilon = 0.1$, $\kappa = 0.01$: theoretical curves (lines) and simulation results (dots).

Furthermore, Fig. 6 illustrates the impact of ϵ and κ on the covert capacity under a finite slot length $L = 302$. The covert capacities of proposed schemes increase with ϵ. For a given ϵ, the covert capacity increases slightly as κ increases, as the joint reception over M frames mitigates the communication disruption caused by bursty strong interference, thereby reducing the impact of the outage probability constraint on covert capacity. As illustrated in Fig. 7, the covert capacity $C_{\text{cov}}(M, L)$ decreases as the minimum elevation angle $\zeta_{\min}$ increases. This degradation occurs because a higher $\zeta_{\min}$ reduces the number of accessible satellites N_A, thereby diminishing directional uncertainty, as shown in (40) and (44). Moreover, as $\zeta_{\min}$ increases, the decreasing trend of $C_{\text{cov}}(M, L)$ gradually slows down. Meanwhile, the covert capacity $C_{\text{cov}}(L)$ of the single-satellite scheme remains unaffected, since it does not exploit directional uncertainty.

The covert capacity with different numbers of satellites N is given in Fig. 8. As N increases, the covert capacity of the single-Sat scheme initially increases and then gradually decreases. The decline occurs because the growing aggregate interference has a greater detrimental impact on Bob's communication link than on Willie's detection capability. However, for the multi-Sat scheme, the covert capacity continues to rise with increasing N, as the directional uncertainty in the transmitting satellite, introduced by the accessible satellite set

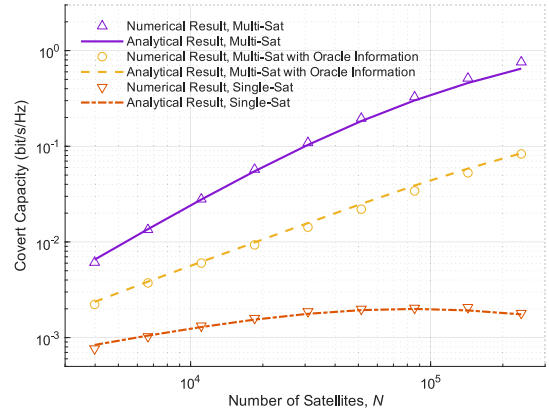

Fig. 9. Covert capacity versus number of satellites N under $L \to \infty$, $\epsilon = 0.1$, $\kappa = 0.01$: theoretical curves (lines) and simulation results (dots).

with $N_A = \rho_A N$ satellites, further degrades Willie's detection capability with the increase of N. With oracle information, the covert capacity initially increases slowly and subsequently accelerates, the reason for which was discussed in Section V-C. In brief, this behavior results from the varying relationship between $E\{\sqrt{M_C}\}$ and N across different intervals. As illustrated in Fig. 9, compared with the case of finite $L = 302$, the covert capacities of all three schemes decrease, while their variation trends with respect to N remain unchanged. Specifically, the single-satellite scheme initially increases and then decreases, whereas the multi-satellite schemes exhibit a consistently increasing trend.

VII. Conclusion

In this paper, we have studied the multi-satellite cooperative covert SatCom scheme in ultra-dense LEO satellite systems, leveraging both aggregate interference power uncertainty and directional uncertainty. Specifically, multiple accessible satellites have cooperated to transmit signals to a legitimate receiver, assisted by aggregate interference generated by sidelobe leakages from other satellite transmissions. Based on the multi-tier satellite interference model, we have quantified the power distribution of the aggregate interference and approximated it using a Gamma distribution. Then, we have analyzed the covert transmission problem for both the single-satellite and multi-satellite cases, and derived expressions for detection probability and covert capacity. Finally, numerical simulations are performed to demonstrate that a positive covert capacity can be achieved in the proposed scheme. Moreover, results have revealed that the covert capacity is significantly enhanced compared to the single-satellite scheme and increases monotonically with the number of satellites N.

Appendix

A. Proof of Theorem 1

Leveraging the independence of satellite distributions across different spherical shells, the mean and the variance of I_{ag} can be derived according to (10),

$$\bar{I}_{\text{ag}} = \sum_{k=1}^{K} E\{N_{I,k}\} E\{I_{k,0}\}, \sigma_I^2 = \sum_{k=1}^{K} \text{Var}\left(\sum_{n \in \mathcal{S}_{I,k}} I_n\right), \tag{52}$$

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where $\mathrm{Var}\left(\sum_{n\in\mathcal{S}_{I,k}} I_n\right) = E\{N_{I,k}\}\mathrm{Var}(I_{k,0}) + \mathrm{Var}(N_{I,k})E\{I_{k,0}\}^2$, $I_n = G'_R(\alpha_n)G'_I(\beta_n)r_n^{-\alpha}P_I|h_n|^2$, and $I_{k,0}$ is a random variable that is identically distributed with I_n for $n \in \mathcal{S}_{I,k}$. According to Corollary 1, the first and second moments of $N_{I,k}$ can be calculated as $E[N_{I,k}] = N_k p_{I,k}$, $\mathrm{Var}[N_{I,k}] = N_k p_{I,k}(1-p_{I,k})$, where $N_k = \rho_k N$. Therefore, the mean and variance of I_{ag} can be calculated as $\bar{I}_{\mathrm{ag}} = N\sum_{k=1}^{K}\rho_k p_{I,k}E\{I_{k,0}\} \triangleq \mu_I N$, and $\sigma_I^2 = N\sum_{k=1}^{K}\left(\rho_k p_{I,k}\mathrm{Var}(I_{k,0}) + \rho_k p_{I,k}(1-p_{I,k})E\{I_{k,0}\}^2\right) \triangleq \nu_I N$, where μ_I and ν_I can be obtained through numerical methods. Note that outlier processing is necessary to prevent sporadic strong interference from distorting the approximation of the aggregate interference power distribution.

B. Proof of Lemma 2

Building on the Gamma distribution approximation of the product of two Gamma random variable [40], we similarly approximate the product of a Gamma distribution and a Gamma with offset as a Gamma distribution with an offset through the method of moments, i.e, $P_{\mathrm{IN}} = (I_{\mathrm{ag}} + \sigma_W^2)V \approx G_T + \sigma_W^2$, where $G_T \sim \mathrm{Gamma}(k_T, \theta_T)$, where $V \sim \mathrm{Gamma}(L, 1/L)$, $E\{V\} = 1$, and k_T, θ_T are calculated by method of moments,

$$k_T\theta_T = E\{G_T\} = E\{P_{\mathrm{IN}}\} - \sigma_W^2 = \bar{I}_{\mathrm{ag}} = \mu_I N \tag{53}$$

$$k_T\theta_T^2 = \mathrm{Var}(G_T) = \frac{(L+1)\nu_I N + (\mu_I N + \sigma_W^2)^2}{L} \tag{54}$$

Therefore, $\eta_{\min}(L)$ is approximated,

$$\begin{aligned}\eta_{\min}(L) &\approx 1 - \max_{\gamma}\left\{\mathbb{P}(\gamma - \sigma_W^2 - P_{W,0} \le G_T \le \gamma - \sigma_W^2)\right\}\\ &= 1 - \max_{\gamma\ge 0}\int_{\gamma}^{\gamma+P_{W,0}} f_T(x)dx,\end{aligned} \tag{55}$$

where $f_T(x)$ denotes the PDF of G_T. For $0 \le k_T \le 1$, we have $\eta_{\min}(L) = 1 - \int_0^{P_{W,0}} f_T(x)dx$, since $f_T(x)$ is monotonically decreasing. For $k_T > 1$, $f_T(x)$ reaches its maximum at $x = (k_T - 1)\theta_T$, and hence $\eta_{\min}(L) > 1 - P_{W,0}\max\{f_T(x)\}$. In summary, $\eta_{\min}(L)$ can be expressed as follows,

$$\eta_{\min}(L) \ge \begin{cases}\dfrac{\Gamma(k_T, P_{W,0}/\theta_T)}{\Gamma(k_T)}, & 0 \le k_T \le 1,\\ 1 - \dfrac{P_{W,0}(k_T-1)^{k_T-1}}{\theta_T e^{k_T-1}\Gamma(k_T)}, & k_T > 1.\end{cases} \tag{56}$$

When N and L are sufficiently large, the Gamma function with a significant large $k_T = \frac{\mu_I^2 N^2 L}{\nu_I NL + \nu_I N + (\mu_I N + \sigma_W^2)^2}$ can be calculated by using Stirling's approximation, i.e., $\Gamma(k_T) = (k_T-1)\Gamma(k_T-1) \approx \sqrt{2\pi}e^{-(k_T-1)}(k_T-1)^{k_T-1/2}$. In general, when $k_T \ge 10$, the approximation error becomes sufficiently small. The approximate lower bound of $\eta_{\min}(L)$ is obtained,

$$\begin{aligned}\eta_{\min}(L) &\ge 1 - \frac{P_{W,0}(k_T-1)^{k_T-1}e^{-(k_T-1)}}{\theta_T\sqrt{2\pi}e^{-(k_T-1)}(k_T-1)^{k_T-1/2}}\\ &= 1 - \frac{P_{W,0}}{\sqrt{2\pi(k_T-1)\theta_T^2}}\\ &\approx 1 - \frac{P_{W,0}}{\sqrt{2\pi k_T\theta_T^2}}.\end{aligned} \tag{57}$$

C. Proof of Lemma 4

Since the sum of independent and identically distributed Gamma random variables also follows a Gamma distribution, $P_{\mathrm{IN,avg}} = \frac{1}{M}\sum_{m=1}^{M} P_{\mathrm{IN},m}$ can also be approximated as $P_{\mathrm{IN,avg}} - \sigma_W^2 \sim \mathrm{Gamma}(k_T^{(M)}, \theta_T^{(M)})$, where $k_T^{(M)}, \theta_T^{(M)}$ are calculated by the method of moments, and $f_T^{(M)}(x)$ is the corresponding PDF. Therefore, $\eta_{\min}(M, L)$ is calculated as follows,

$$\begin{aligned}\eta_{\min}(M,L) &\approx 1 - \max_{\gamma>0}\sum_{b=0}^{M} p_B(b)\int_{\gamma}^{\gamma+b\frac{P_{W,0}}{M}} f_T^{(M)}(x)dx\\ &\ge 1 - \sum_{b=0}^{M} p_B(b)\max_{\gamma>0}\int_{\gamma}^{\gamma+b\frac{P_{W,0}}{M}} f_T^{(M)}(x)dx\end{aligned} \tag{58}$$

where $p_B(b)$ denotes the probability that $B = b$. When N and L are sufficiently large, $\eta_{\min}(M, L)$ can be approximated as

$$\begin{aligned}&\eta_{\min}(M,L)\\ &\overset{(a)}{\ge} 1 - \sum_{b=0}^{M} p_B(b)\max\{f_T^{(M)}(x)\}b\frac{P_{W,0}}{M}\\ &\overset{(b)}{=} 1 - \frac{P_{W,0}(k_T^{(M)}-1)^{k_T^{(M)}-1}e^{-(k_T^{(M)}-1)}}{N_A\theta_T^{(M)}\Gamma(k_T^{(M)})}\\ &\overset{(c)}{\ge} 1 - \frac{P_{W,0}(k_T^{(M)}-1)^{k_T^{(M)}-1}e^{-(k_T^{(M)}-1)}}{N_A\theta_T^{(M)}e^{-(k_T^{(M)}-1)}(k_T^{(M)}-1)^{(k_T^{(M)}-1/2)}\sqrt{2\pi}}\\ &\approx 1 - \frac{P_{W,0}}{\sqrt{2\pi}N_A\sqrt{k_T^{(M)}(\theta_T^{(M)})^2}},\end{aligned} \tag{59}$$

where (a) follows from the fact that the maximum integral of $f_T^{(M)}(x)$ over an interval is upper-bounded by the interval length multiplied by the maximum of $f_T^{(M)}(x)$, (b) holds since $\sum_{b=0}^{M} p_B(b)b = E\{B\} = \frac{M}{N_A}$, and (c) is obtained by applying Stirling's approximation.

D. Proof of Theorem 4

According to (48), $T_W^{\mathrm{ora}}(M, L)$ is calculated as follows,

$$\mathbf{H}_0 : T_W^{\mathrm{ora}}(M,L) = \frac{1}{M_C}\sum_{m\in\mathcal{M}_C} P_{\mathrm{IN},m}, \tag{60}$$

$$\begin{aligned}&\mathbf{H}_1 : T_W^{\mathrm{ora}}(M,L)\\ &= \frac{1}{M_C}\sum_{m\in\mathcal{M}_C} P_{\mathrm{IN},m} + P_{W,0}\frac{1}{M_C}\sum_{m\in\mathcal{M}_C}\delta_{s_W,s_m}\\ &= \frac{1}{M_C}\sum_{m\in\mathcal{M}_C} P_{\mathrm{IN},m} + P_{W,0}.\end{aligned} \tag{61}$$

Since the satellite s_m is randomly selected from the N_A accessible satellites to transmit information in each slot m, and Willie selects a fixed satellite (direction) to monitor, the size M_C of the successfully monitored signal set $\mathcal{M}_C$ follows a binomial distribution, i.e., $M_C \sim \mathrm{Binomial}\left(M, \frac{1}{N_A}\right)$, where $\frac{1}{N_A}$ represents the probability of successful

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detection in a single attempt. Substituting (60) and (61) into (9),

$$\begin{aligned}&\eta_{\min}^{\text{ora}}(M,L)\\&\overset{(a)}{=}\sum_{M_C=0}^{M}\eta_{\min}^{\text{ora}}(M,L|M_C)\mathbb{P}(M_C)\\&\overset{(b)}{=}1-\sum_{M_C=1}^{M}\max_{\gamma}\mathbb{P}(\gamma-P_{W,0}\le P_{\text{IN,avg}}^{\text{ora}}\le\gamma)\mathbb{P}(M_C),\end{aligned}\tag{62}$$

where $P_{\text{IN,avg}}^{\text{ora}}=\frac{1}{M_C}\sum_{m\in\mathcal{M}_C}P_{\text{IN},m}$, $\mathbb{P}(M_C)=\left(\frac{1}{N_A}\right)^{M_C}\left(1-\frac{1}{N_A}\right)^{M-M_C}$, (a) is obtained as that Willie knows which slots were successfully received in advance and thus also knows the number M_C of terms, and (b) is obtained as $\eta_{\min}^{\text{ora}}(\mathcal{M}_C,L|0)=1$. $P_{\text{IN,avg}}^{\text{ora}}$ can also be approximated as $P_{\text{IN,avg}}^{\text{ora}}-\sigma_W^2\sim\text{Gamma}(k_T^{(C)},\theta_T^{(C)})$, where $k_T^{(C)}\theta_T^{(C)}=\mu_I N$ and $k_T^{(C)}(\theta_T^{(C)})^2=\frac{\nu_I N}{M_C}+\frac{\nu_I N+(\mu_I N+\sigma_W^2)^2}{M_C L}$. Therefore, $\eta_{\min}^{\text{ora}}(M,L)$ is calculated as follows,

$$\begin{aligned}&\eta_{\min}^{\text{ora}}(M,L)\\&\approx 1-\sum_{M_C=1}^{M}\max_{\gamma>0}\int_{\gamma}^{\gamma+P_{W,0}}\frac{x^{k_T^{(C)}-1}e^{-x/\theta_T^{(C)}}}{(\theta_T^{(C)})^{k_T^{(C)}}\Gamma(k_T^{(C)})}dx\mathbb{P}(M_C)\\&\ge 1-\sum_{M_C=1}^{M}\frac{P_{W,0}(k_T^{(C)}-1)^{k_T^{(C)}-1}e^{-(k_T^{(C)}-1)}}{\theta_T^{(C)}\Gamma(k_T^{(C)})}\mathbb{P}(M_C)\\&\approx 1-\sum_{M_C=1}^{M}\frac{P_{W,0}}{\sqrt{2\pi}\sqrt{\frac{\nu_I N}{M_C}+\frac{\nu_I N+(\mu_I N+\sigma_W^2)^2}{M_C L}}}\mathbb{P}(M_C)\\&=1-\frac{E\{\sqrt{M_C}\}P_{W,0}}{\sqrt{2\pi\left(\nu_I N+\frac{\nu_I N+(\mu_I N+\sigma_W^2)^2}{L}\right)}}\ge 1-\epsilon.\end{aligned}\tag{63}$$

The received signal power of Willie is obtained as follows,

$$P_{W,0}\le\frac{\epsilon\sqrt{2\pi\left(\nu_I N+\frac{\nu_I N+(\mu_I N+\sigma_W^2)^2}{L}\right)}}{E\{\sqrt{M_C}\}}.\tag{64}$$

The outage probability analysis is similarly to (43) in Theorem 3, and $C_{\text{cov}}^{\text{ora}}(M,L)$ is obtained by substituting (43) and (64) into (39).

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